



NITI Aayog



SECTOR AT A GLANCE 09

# Atomic Energy in India

Energy in India

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# ATOMIC ENERGY IN INDIA

ENERGY IN INDIA

The '**Sector-at-a-Glance**' provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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# Overview

Atomic energy, which involves harnessing energy from atomic or nuclear reactions, presently constitutes 1.5% of India's energy mix.<sup>1</sup> As per IESS estimates, this is expected to increase to 3.6% in 2047.<sup>2</sup> Presently, there are 22 nuclear reactors in operation, and 21 nuclear reactors are at various stages of development, with nine expected to be commissioned by 2024-25.<sup>3</sup>

India has a significant opportunity to expand its atomic energy sector to wide applications. Nuclear power prevents the release of 41 million tonnes of CO<sub>2</sub> emissions annually in India when compared to the emissions that would result from generating equivalent electricity using coal-based thermal power plants.<sup>4</sup> Along with energy generation, atomic energy in India is being used to treat cancer, sterilise medical products, and increase the shelf life of agricultural products like onions and mangoes.

Despite its advantages, the potential of atomic energy is not fully utilized, and it faces a range of challenges related to negative public perception, restrictive private sector participation, high cost of production, and project delays and cost overruns.

Given these opportunities and challenges, increasing private sector participation in atomic power generation, technological innovation for cost-reduction, and increasing domestic production of uranium are the major policy focus areas. Efforts including continued nuclear cooperation with other countries, focusing on technology exchange, and fuel supply assurance aim to bolster India's energy independence and harness nuclear technology to provide a safer and higher quality of life for its citizens.

Accelerating India's low-carbon transition through better integration of nuclear energy, ensuring safe use and operations, advancing innovative technologies like small modular reactors, and enhancing effective waste management are critical to advancing India's atomic energy ambitions. This note presents key trends related to different components of atomic energy in India and actions for the way forward.

Figure 1: Key atomic energy statistics



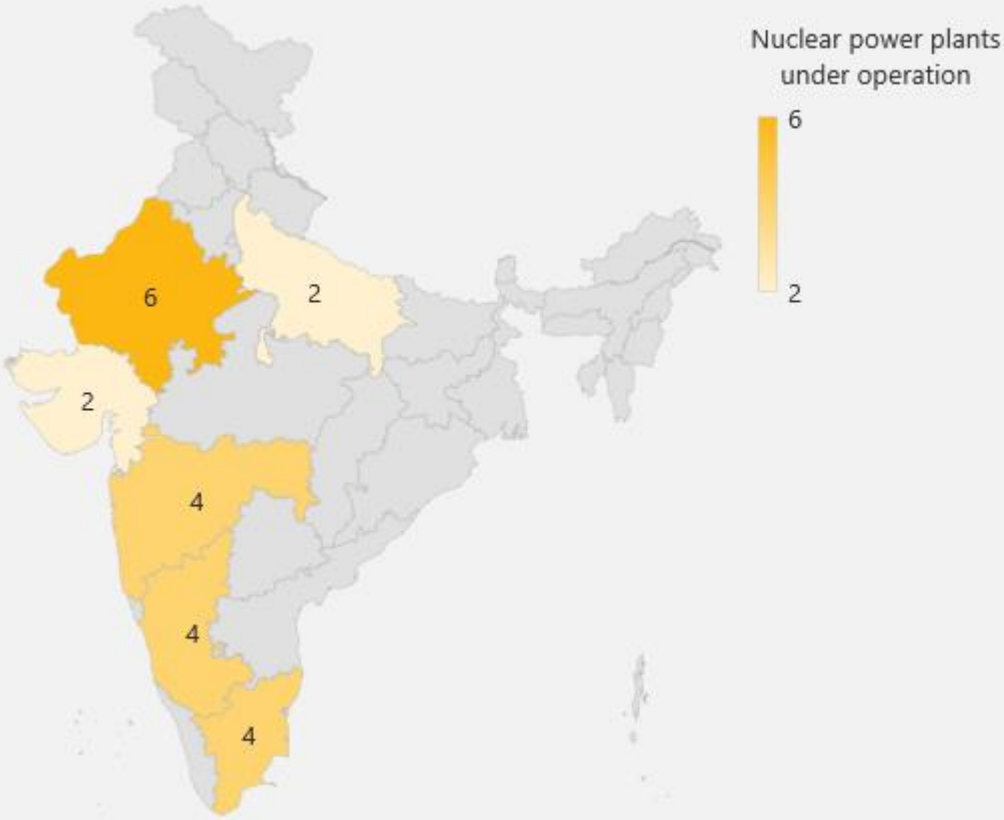
<sup>1</sup> NITI Aayog, India Energy Security Scenario, accessed December 2023

<sup>2</sup> NITI Aayog, India Energy Security Scenario, accessed December 2023

<sup>3</sup> Department of Atomic Energy, Proposals for New Atomic Power Plants, 2019

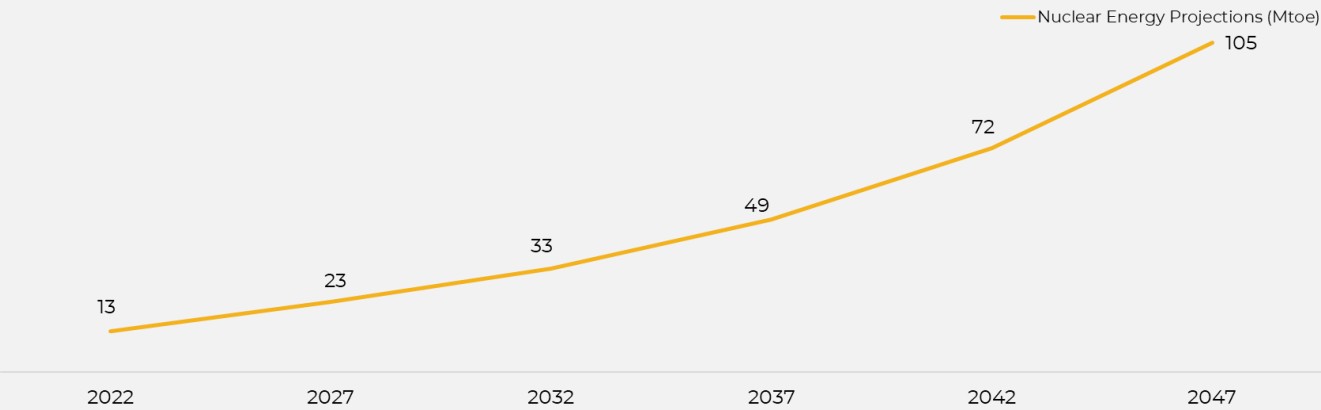
<sup>4</sup> PIB, Union Minister Dr. Jitendra Singh says, presently (2020-21) nuclear power is saving 41 million tonnes of CO<sub>2</sub> emissions annually, compared to the emissions that would be generated by equivalent electricity generation from coal based thermal power plants, 2023

Figure 2: List of nuclear power plants under operation in India (2023)



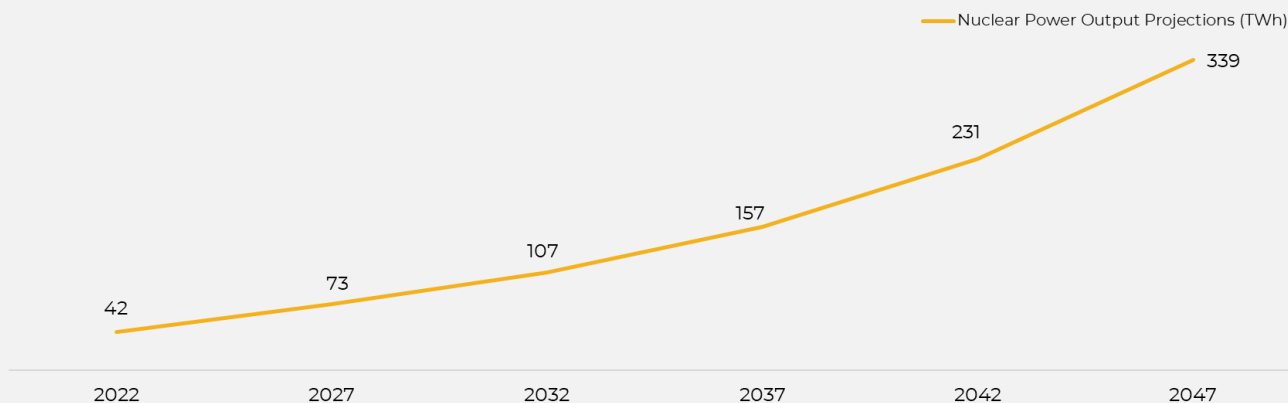
Source: Atomic Energy Regulatory Board

Figure 3: Current and expected projections for atomic energy in India's energy mix (2022-2047), in Mtoe



Source: NITI Aayog, India Energy Security Scenario, accessed December 2023

Figure 4: Current and expected power output projections for atomic energy in India's electricity mix (2022-2047), in terawatt hours



Source: NITI Aayog, India Energy Security Scenario, accessed December 2023

# Opportunities and challenges

## Opportunities

This section highlights the outcomes policymakers can achieve by expanding the installed capacity of atomic energy. These outcomes include innovative application in sectors such as agriculture and healthcare, productive use of available thorium reserves, and contributing to India's clean energy transition goals.



### **Support India's clean energy transition goals:**

India has set an ambitious target to achieve net-zero emissions by 2070. Atomic energy is a clean energy source, and its expansion can help India meet its ambitious targets and ensure sustainable long-term security for the country.<sup>5</sup>



### **Take a globally leading role in the safe and sustainable development of nuclear energy:**

India possesses approximately 25% of the world's thorium reserves, a crucial raw material for producing atomic energy. Notably, it is the sole developing nation to develop and deploy indigenous atomic reactors for electricity generation successfully.<sup>6</sup> Leveraging its robust research and development capabilities, India is currently bridging the global gap in commercial thorium reactor programs by making substantial investments in nuclear technology R&D<sup>7</sup>. Such efforts could propel India to the forefront of scaling viable solutions for worldwide adoption. India's continued investments in atomic energy towards setting up new nuclear reactors and incentivising domestic manufacturing of essential equipment can make it a prominent global leader in nuclear power production.<sup>8</sup>



### **Innovative and peaceful applications in agriculture and healthcare sectors:**

India has pioneered successful applications of atomic energy across diverse sectors, such as in agriculture, to extend the shelf life of fruits and food products and in healthcare for diagnostics imaging, advanced cancer treatment through radiation therapy, and nuclear medicine.<sup>9</sup>



### **Provision for Base Load Power:**

While India is making substantial strides in the adoption of renewable energy, targeting around 300-380 GW of solar and wind capacity by 2030, the weather-dependent nature of these resources renders them unsuitable for consistent base load support.<sup>10</sup> Nuclear power, on the other hand, stands as a clean and environment-friendly base-load source of electricity generation that is available 24X7. With immense potential, it offers the country long-term energy security in a sustainable manner. The expansion of nuclear power capacity is poised to play a key role in the country's energy transition, aligning with the goal of achieving a net-zero economy.<sup>11</sup>

5 PIB – Department of Atomic energy – Government of India, Use atomic energy for power generation, 2022; International Energy Agency, India's clean energy transition is rapidly underway, benefiting the entire world, 2022

6 Bhabha Atomic Research Centre, BARC activities for Indian Nuclear Power Program, Accessed July 2023

7 Press Information Bureau, Thorium-Based Nuclear Reactors, 2019

8 Department of Atomic Energy, Union Minister Dr Jitendra Singh says, nearly 9% share of electricity likely to be contributed from India's nuclear sources by the year 2047; Department of Atomic Energy,

9 Department of Atomic Energy, Use of Atomic energy, 2018; Department of Atomic Energy, Union Minister Dr Jitendra Singh says, nearly 9% share of electricity likely to be contributed from India's nuclear sources by the year 2047, 2023

10 Central Electricity Authority, Ministry of Power, Government of India, Report on Optimal Generation Capacity Mix for 2029-2030, 2023

11 Press Information Bureau, Use of Nuclear Energy for Power Generation, 2022



### **Develop Small Modular Reactors (SMRs):**

SMRs are a promising technology for industrial decarbonization, particularly in situations requiring a reliable and continuous power supply. SMRs with up to 300 MW capacity offer flexibility in design and are safer than large nuclear plants. SMRs offer significant cost savings and reduced construction time since they may be factory-built unlike the conventional nuclear reactors that are built on-site. Exploring the participation of the private sector and startups in the development of this critical technology within India is essential.<sup>12</sup>

## **Challenges**

This section highlights some of the major challenges and constraints facing atomic energy in India. Efforts to resolve such challenges will ensure India effectively harness the potential of its existing reserves to further its energy security and clean energy goals.



### **Public perception and its impact on the large-scale deployment of atomic energy:**

One of the most significant concerns is the association of atomic energy with nuclear disasters, such as Chernobyl and Fukushima, which have left lasting impressions of the catastrophic consequences of nuclear power on the environment and public health.<sup>13</sup> Other concerns include fears that it fosters nuclear weapons proliferation, the inherent safety of nuclear reactors, nuclear waste disposal, and the potential dangers of radiation to human health and well-being.<sup>14</sup> Such fears increase distrust in the public and affect the country-wide development and deployment of atomic energy technologies. For instance, the Kudankulam nuclear power plant faced a delay of over six months due to local protests.<sup>15</sup>



### **Restrictions on private sector participation:**

The private sector's role in nuclear power generation is primarily confined to equipment manufacturing and providing support services. However, as the demand for atomic energy grows, and with budgetary constraints in mind, augmenting private sector participation becomes imperative.<sup>16</sup> The NITI Aayog has suggested permitting foreign investment in the nation's nuclear power sector and increasing involvement from domestic private companies.<sup>17</sup>



### **High cost of production and limited capital for atomic energy:**

The cost of installing domestically designed and built nuclear power units is about INR 16 Crores/MW.<sup>18</sup> This cost is comparable to large hydro-power projects or liquefied natural gas (LNG) liquefaction plants. However, raising capital for nuclear power plants is more challenging due to their unique risks, such as low probability but high-risk events like natural disasters and nuclear proliferation concerns. Additionally, long lead times, potential construction problems, delays, cost overruns, and the uncertainty of technological changes increase costs associated with atomic energy.<sup>19</sup>



### **Need for optimal utilisation of domestic uranium resources:**

Uranium is an essential fuel for Pressurised Water Reactors (PWR). Most of India's domestic production of Uranium happens in the Jaduguda mines of Jharkhand. However, these mines are now one of the deepest sites in the world, where extraction costs are high, making it easier

<sup>12</sup> Department of Atomic Energy, Union Minister Dr Jitendra Singh says, India taking steps for development of Small Modular Reactors (SMR), with up to 300 MW capacity to fulfill its commitment to Clean Energy transition, 2022

<sup>13</sup> Department of Atomic Energy, Public Perception about Atomic Energy – Myths Vs Realities, Accessed July 2023

<sup>14</sup> Department of Atomic Energy, Public Perception about Atomic Energy – Myths Vs Realities, Accessed July 2023

<sup>15</sup> International Atomic Energy Agency, Challenges in developing Nuclear Power Infrastructure, 2017; Department of Atomic Energy, Target for Atomic Energy, 2017

<sup>16</sup> Observer Research Foundation, India's targets for atomic energy: Moving closer?, 2023

<sup>17</sup> Observer Research Foundation, India's targets for atomic energy: Moving closer?, 2023

<sup>18</sup> Government of India, India's Atomic Energy Programme, 2017

<sup>19</sup> Observer Research Foundation, India's targets for atomic energy: Moving closer?, 2023

to import Uranium.<sup>20</sup> The Committee on Estimates, in its report on "Import of Uranium for Nuclear Plants", has highlighted the need for new domestic mines in underutilised reserves across Meghalaya, Andhra Pradesh, Karnataka, Tamil Nadu, and Haryana to increase domestic production of Uranium.<sup>21</sup>



### **High cost of long-term decommissioning and disposal of nuclear waste:**

India produces approximately 4 tonnes of nuclear waste per gigawatt-year of electricity generated.<sup>22</sup> The safe disposal of radioactive waste, including uranium tailings, transuranic waste, and used fuel rods, comes at a significant cost. The government's plan to involve private companies in atomic energy production through Small Modular Reactors (SMRs) also faces challenges because of the expenses related to waste management and long-term decommissioning responsibilities.<sup>23</sup>

20 Department of Atomic Energy, Import of Uranium for Nuclear Plants - Action taken by Government on the Recommendations/Observations contained in the Thirty-First Report (16th Lok Sabha) of the Committee on Estimates, 2021

21 Department of Atomic Energy, Import of Uranium for Nuclear Plants - Action taken by Government on the Recommendations/Observations contained in the Thirty-First Report (16th Lok Sabha) of the Committee on Estimates, 2021

22 Lok Sabha Answer by the Minister of State For Personnel, Public Grievances & Pensions And Prime Minister's Office (Dr. Jitendra Singh), Nuclear Waste Disposal, 2014

23 OECD, The Financing of Nuclear Power Plants, 2009

# Institutional framework and policy landscape —

## Key ministries and departments

This section highlights the major central government bodies and national-level institutions that oversee atomic energy.



### NITI Aayog

The NITI Aayog serves as the apex public policy think tank of the Government of India. Within the energy vertical, deals with five key sectors: power, coal, petroleum and natural gas, new and renewable energy, and atomic power. It conducts policy analysis of the atomic energy sector; and undertakes steps to improve energy planning, and data management in India.



### Department of Atomic Energy (DAE)

DAE is responsible for the entire spectrum of activities related to nuclear science and technology. This encompasses construction and operation of nuclear reactors, research and development of nuclear technologies, and exploration of atomic minerals.



### Atomic Energy Regulatory Board

The board performs specific regulatory and safety functions in accordance with the Atomic Energy Act of 1962. It conducts reviews of nuclear and industrial safety aspects within nuclear facilities, formulates safety policies and codes, and ensures compliance with its regulations.

## Major policies and programs

This section highlights existing policies related to atomic energy and their most prominent features. A more comprehensive list of policies is provided in the policy repository of the NITI Knowledge Platform.



### Three-stage nuclear power program:

The program was set up in the 1950s to utilise indigenous Uranium and thorium reserves to meet the country's energy needs and promote self-sufficiency in nuclear power generation. It involves using natural Uranium in Pressurised Heavy Water Reactors (PHWRs) as the first stage, followed by utilising Plutonium from PHWR spent fuel in Fast Breeder Reactors (FBRs) and deploying advanced nuclear reactors fuelled primarily by thorium as the third stage.<sup>24</sup>



### Amendments to the Atomic Energy Act 1962:

The 2015 amendments to the Atomic Energy Act allowed India's Nuclear Power Corporation (NPCIL) to partner with public companies like NTPC, IOCL, and NALCO on nuclear power projects. This aims to boost domestic investment and technology sharing in the atomic energy sector.<sup>25</sup>



### Nuclear cooperation agreements with other countries:

The Government of India has signed multilateral agreements with 18 countries to facilitate cooperation in the peaceful uses of atomic energy. These agreements can assist India in boosting technology transfer, advanced research and development, workforce training, and fuel supply assurance.<sup>26</sup>

<sup>24</sup> Department of Atomic Energy, Thorium-based nuclear reactors, 2019

<sup>25</sup> Department of Atomic Energy, Investment in Atomic Energy, 2020

<sup>26</sup> Department of Atomic Energy, Collaboration for Developing Atomic Energy, 2022; ORF, India's civil nuclear agreements: A new dimension in India's global diplomacy, 2019



### **Streamlined regulatory processes and high safety standards:**

India has implemented strict regulations and standards to ensure nuclear power plants' safe functioning. The AERB's role has been strengthened to ensure safety and regulatory compliance. As a result, India's nuclear program is one of the safest in the world, with no severe security incident in the past five decades.<sup>27</sup>

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<sup>27</sup> Ministry of External Affairs, Nuclear Security in India, Accessed June 2023

# Way forward

This section highlights considerations for policymakers seeking to advance policy formulation and implementation in this sector. These are based on reports and materials prepared by key government bodies and expert institutions like the IAEA and the AERB.

|                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p><b>Promote a clean energy transition</b></p>                            | <p><b>Invest in renewable integration</b></p> <ul style="list-style-type: none"> <li>- Develop policies to integrate nuclear power with renewable energy sources (such as solar and wind) to create a balanced and reliable energy mix</li> </ul> <p><b>Enhance grid stability</b></p> <ul style="list-style-type: none"> <li>- Enable and support DISCOMS to ensure stable grid management through energy storage technologies and improved transmission infrastructure, allowing seamless integration of nuclear energy into the existing power infrastructure</li> </ul> <p><b>Incentivize hybrid systems</b></p> <ul style="list-style-type: none"> <li>- Encourage hybrid power plants that combine nuclear and renewable energy technologies for efficient and continuous electricity generation</li> </ul>                                                                                        |
|  <p><b>Ensure safe and sustainable nuclear energy operations</b></p>       | <p><b>Strengthen regulatory framework</b></p> <ul style="list-style-type: none"> <li>- Continue advancing robust safety regulations and oversight mechanisms (including more frequent on-site inspections, additional independent verifications, and stronger emergency arrangements) to prevent accidents and ensure the safe operation of nuclear facilities</li> </ul> <p><b>Invest in research and development</b></p> <ul style="list-style-type: none"> <li>- Allocate resources for advanced reactor designs, materials, and fuel cycles that enhance safety, reduce waste, and improve efficiency</li> </ul> <p><b>Promote public awareness</b></p> <ul style="list-style-type: none"> <li>- Continue educating the public about the safety features of nuclear power and address misconceptions through transparent communication on the safe handling and disposal of nuclear waste</li> </ul> |
|  <p><b>Fostering innovation through small modular reactors (SMRs)</b></p> | <p><b>Facilitate SMR deployment</b></p> <ul style="list-style-type: none"> <li>- Create a supportive ecosystem for SMR development, including streamlined licensing processes and financial incentives</li> </ul> <p><b>Collaborate with industry</b></p> <ul style="list-style-type: none"> <li>- Partner with private companies and research institutions to accelerate SMR research, testing, and commercialization</li> </ul> <p><b>Demonstrate pilot projects</b></p> <ul style="list-style-type: none"> <li>- Invest in pilot SMR projects to showcase their viability, safety, and economic benefits</li> </ul>                                                                                                                                                                                                                                                                                   |
|                                                                           | <p><b>Enhance the implementation of closed fuel cycle strategies</b></p> <ul style="list-style-type: none"> <li>- Develop policies to reprocess spent nuclear fuel, extract valuable isotopes, and minimize waste volume</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

**Implement  
effective  
waste  
management  
and resource  
optimization**

**Explore thorium-based reactors**

- Investigate thorium-based reactors as an alternative fuel cycle to optimize uranium resources

**Encourage international cooperation**

- Collaborate on R&D and share best practices with other countries on nuclear safety, radiation protection and monitoring, decontamination and dismantling of nuclear facilities, and radioactive waste management technologies

## Recommended readings

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